

Relational Kernels: Geometry, Bell Nonlocality, and the Exact Noisy-Tsirelson Completion Curve

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Updated theorem draft - incorporating machine-checked noisy-Tsirelson completion theorem

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Abstract

We study normalized positive semidefinite (PSD) relational kernels as common-origin structures from which geometric and Bell-correlation properties may be derived. A symmetric normalized PSD kernel G can define a candidate geometry through $d(i,j) = -\log |G(i,j)|$, while selected cross-sector entries define CHSH correlators. Geometric consistency is encoded by μ -submultiplicativity, $|G(i,k)| \geq \mu |G(i,j)| |G(j,k)|$, which is equivalent to a triangle-inequality defect for the log-distance. The core no-go theorem states that exact Tsirelson saturation is incompatible with any nonzero uniform geometric consistency on the Bell sector. This update adds an exact, machine-checked theorem for the canonical noisy-Tsirelson family: among all PSD completions compatible with $E_{00}=E_{01}=E_{10}=x$ and $E_{11}=-x$, the maximal compatible geometric consistency is $\mu_{\max}(x) = \min(1, (\sqrt{1-x^2} - x)/x^2)$. Thus, along the standard path toward maximal quantum CHSH violation, the amount of compatible relational geometry is exactly computable and collapses continuously to zero at the Tsirelson point. We also introduce the observable completion invariant $\mu_{\max}(E)$, which depends only on experimentally accessible Bell correlators, and prove an exponential-decay bridge theorem showing that approximate exponential metric decay implies positive μ -submultiplicativity. The full global tradeoff $C^*(\mu)$ remains open, but the noisy-Tsirelson path is now solved exactly.

1. Introduction

Bell nonlocality shows that quantum correlations can violate classical local-hidden-variable constraints. In the CHSH scenario, classical correlations satisfy $|S| \leq 2$, quantum correlations satisfy $|S| \leq 2\sqrt{2}$, and no-signaling correlations can reach $|S| = 4$. A central question is what structural features distinguish quantum correlations from unconstrained nonlocal correlations.

This paper studies normalized PSD kernels as a common relational substrate. A PSD kernel is equivalently a Gram matrix of unit vectors. The same matrix may be read in two ways: as a similarity kernel inducing a log-distance $d(i,j) = -\log |G(i,j)|$, and as a source of Bell correlators $E(a,b) = G(A_a, B_b)$.

The central consistency condition is μ -submultiplicativity: $|G(i,k)| \geq \mu |G(i,j)| |G(j,k)|$. This condition says that if i is strongly related to j and j is strongly related to k , then i and k cannot be arbitrarily unrelated. When $\mu=1$, this is exactly the triangle inequality for d . When $0 < \mu < 1$, it gives an approximate triangle inequality with additive defect $-\log \mu$.

The original theorem showed that exact Tsirelson saturation forces a breakdown of this transitivity condition on the Bell sector. The new result goes further: for the canonical noisy-Tsirelson path, the maximal compatible value of μ is computed exactly. This turns the spectrum between geometry and nonlocality into an explicit closed-form curve in a physically meaningful one-parameter family.

Two additions make the framework more operational and physically grounded. First, $\mu_{\max}(E)$ is defined directly from observed Bell correlators by optimizing over PSD completions. Second, an exponential-decay bridge theorem

shows that mu-submultiplicativity naturally appears in systems whose correlations are approximately exponential functions of an underlying metric distance.

2. Relational Kernel Framework

Definition 2.1 (Normalized PSD kernel). A kernel G on a finite index set I is normalized PSD if G is real symmetric, positive semidefinite, and $G(i,i)=1$ for all i in I . Equivalently, there exist unit vectors v_i such that $G(i,j)=\langle v_i, v_j \rangle$.

Definition 2.2 (Geometric consistency). For μ in $[0,1]$, G is μ -submultiplicative on a subset T of indices if $|G(i,k)| \geq \mu |G(i,j)| |G(j,k)|$ for all i,j,k in T .

Definition 2.3 (Intrinsic consistency parameter). For a finite kernel G , define $\mu(G)=\min |G(i,k)|/(|G(i,j)| |G(j,k)|)$, with triples having vanishing denominator treated separately or excluded by threshold in numerical work. This measures the worst-case failure of multiplicative relational transitivity.

The log-distance $d(i,j)=-\log |G(i,j)|$ satisfies $d(i,k) \leq d(i,j)+d(j,k)-\log \mu$ whenever G is μ -submultiplicative. Thus $\delta=-\log \mu$ is a triangle-inequality defect. Large μ corresponds to strong geometric coherence; μ near zero corresponds to severe nontransitivity.

3. Bell Sector and General Obstruction

In the CHSH sector $T=\{A0,A1,B0,B1\}$, define $x_{ab}=G(A_a,B_b)$ and $S=x_{00}+x_{01}+x_{10}-x_{11}$. The standard PSD/Gram formulation gives $|S| \leq 2 \sqrt{2}$. Exact saturation imposes a rigid orthogonality pattern: in the usual Tsirelson extremizer, same-party overlaps $G(A0,A1)$ and $G(B0,B1)$ vanish.

Theorem 3.1 (Geometric obstruction). If a normalized PSD Bell-sector kernel satisfies uniform μ -submultiplicativity with any $\mu>0$, then it cannot attain $|S|=2 \sqrt{2}$. Equivalently, exact Tsirelson saturation is incompatible with nonzero uniform geometric consistency on the Bell sector.

Proof sketch. Tsirelson saturation forces an orthogonality relation such as $G(A0,A1)=0$. Submultiplicativity then gives $|G(A0,A1)| \geq \mu |G(A0,B_b)| |G(B_b,A1)|$. For $\mu>0$ this forces at least one of the two adjacent Bell correlators to vanish for each b , preventing the four-correlator Tsirelson pattern. The machine-checked version of this obstruction appears in the existing CHSH/obstruction Lean bundles.

This theorem is qualitative. It proves that maximal Bell violation and nonzero uniform geometric transitivity cannot coexist on the same four-point sector. It does not by itself determine the sharp quantitative tradeoff $C^*(\mu)$. The next section solves a canonical one-parameter completion problem exactly.

4. PSD-Completion Diagnostic for Observed Bell Data

Real CHSH experiments provide the four cross-correlators $E_{00}, E_{01}, E_{10}, E_{11}$. They do not directly provide same-party Gram entries such as $G(A0,A1)$ and $G(B0,B1)$. We therefore define a completion problem.

$$G(E,y,z) = [[1,z,E00,E01],[z,1,E10,E11],[E00,E10,1,y],[E01,E11,y,1]].$$

Here $z=G(A0,A1)$ and $y=G(B0,B1)$ are free completion variables. We require $G(E,y,z)$ to be PSD, and define

$$\mu_{\max}(E) = \max_{\{y,z: G(E,y,z) \text{ PSD}\}} \mu(G(E,y,z)).$$

This quantity asks: among all normalized PSD Gram completions compatible with the observed CHSH correlators, what is the largest possible geometric consistency? It is therefore a representation-aware diagnostic for Bell data.

4.1 Observable invariant

Definition 4.1 (Observable geometric consistency). For a CHSH correlator table E , define $\mu_{\max}(E)$ to be the supremum of $\mu(G)$ over all normalized PSD completions $G(E,y,z)$ whose cross-party entries agree with E . In finite-dimensional computations this is simply an optimization over y,z in $[-1,1]$ subject to $G(E,y,z)$ being PSD.

The point of this definition is that $\mu_{\max}(E)$ depends only on the experimentally accessible Bell correlators. It asks: among all Gram representations compatible with the observed data, what is the most geometrically coherent relational explanation still possible?

Proposition 4.2 (Observable Tsirelson obstruction). For the Tsirelson-optimal table $E_{00}=E_{01}=E_{10}=1/\sqrt{2}$, $E_{11}=-1/\sqrt{2}$, one has $\mu_{\max}(E)=0$.

Proof sketch. Any normalized PSD completion of the Tsirelson-optimal table saturates the Gram-vector Tsirelson bound. The equality condition forces same-party overlaps such as $G(A_0,A_1)$ and $G(B_0,B_1)$ to vanish while the cross-party correlators remain nonzero. Therefore the ratios controlling μ vanish, so no positive uniform μ is possible.

For real experiments with uncertainty, one can also define an epsilon-robust version $\mu_{\max}^{\epsilon}(E)$ by allowing $|G(A_a,B_b)-E_{ab}| \leq \epsilon_{ab}$. This is the natural quantity to use when fitting finite-sample Bell data.

5. Exact Noisy-Tsirelson Completion Curve

Consider the canonical noisy-Tsirelson family

$$E_{00} = E_{01} = E_{10} = x, \quad E_{11} = -x, \quad 0 \leq x \leq 1/\sqrt{2}.$$

Equivalently, $x=v/\sqrt{2}$, where v in $[0,1]$ is the usual white-noise visibility and $S=4x=2\sqrt{2}v$. The corresponding Bell-sector PSD completion is

$$G(y,z) = [[1,z,x,x],[z,1,x,-x],[x,x,1,y],[x,-x,y,1]].$$

Theorem 5.1 (Exact noisy-Tsirelson completion curve). For the noisy-Tsirelson family above, the maximal geometric consistency parameter compatible with a normalized PSD completion and full Bell-sector μ -submultiplicativity is

$$\mu_{\max}(x) = \min(1, (\sqrt{1-x^2}-x)/x^2).$$

Equivalently, with $x=v/\sqrt{2}$,

$$\mu_{\max}(v) = \min(1, (\sqrt{1-v^2/2}-v/\sqrt{2})/(v^2/2)).$$

5.1 Proof sketch

The proof is a two-variable PSD-completion argument. The determinant of the completed matrix is $\det G = (1-y^2)(1-z^2)-4x^2(1-x^2)$. Since PSD matrices have nonnegative determinant, any PSD completion satisfies $(1-y^2)(1-z^2) \geq 4x^2(1-x^2)$.

Let $m=\min(|y|,|z|)$. Then $1-y^2 \leq 1-m^2$ and $1-z^2 \leq 1-m^2$, so the determinant inequality implies $(1-m^2)^2 \geq 4x^2(1-x^2)$. For $0 \leq x \leq 1/\sqrt{2}$, this gives $m \leq \sqrt{1-x^2}-x$.

The upper bound is sharp. Set $y=z=t=\sqrt{1-x^2}-x$. The theorem engine proves PSD directly via a sum-of-squares decomposition of the quadratic form. For this completion, $\min(|y|,|z|)=t$.

Finally, μ -submultiplicativity implies $\mu x^2 \leq |y|$ and $\mu x^2 \leq |z|$. Therefore $\mu \leq t/x^2$. Conversely, choosing $y=z=t$ and $\mu=\min(1,t/x^2)$ satisfies all ordered triple constraints by finite case analysis over Fin 4. This completes the exact formula.

5.2 Machine-checked theorem artifacts

The theorem engine formalized and proved the following lemmas in RequestProject/NoisyTsirelsonCompletion.lean with no sorry and only standard Lean axioms (propext, Classical.choice, Quot.sound):

det_fin_four - expands the determinant of any 4x4 real matrix into its 24-term Leibniz formula.

noisyTsirelsonGram_det - proves $\det(G)=(1-y^2)(1-z^2)-4x^2(1-x^2)$.

psd_det_upper_bound - derives the PSD determinant inequality.

min_overlap_upper_bound - proves $\min(|y|,|z|) \leq \sqrt{1-x^2}-x$.

noisyTsirelsonGram_achievable_psd - proves PSD for $y=z=\sqrt{1-x^2}-x$.

mu_constraints - proves that μ -submultiplicativity implies $\mu x^2 \leq |y|$ and $\mu x^2 \leq |z|$.

mu_upper_bound - derives the upper bound $\mu \leq (\sqrt{1-x^2}-x)/x^2$.

mu_achievable - proves achievability of $\mu=\min(1,t/x^2)$ and full submultiplicativity by finite case analysis.

6. Visualization and Numerical Confirmation

The exact theorem converts the earlier optimizer sweep from conjectural evidence into a visualization of the closed-form curve. Figure 1 overlays the numerical optimizer values with the exact expression. The curve remains at $\mu=1$ through the low-correlation regime, then declines continuously and reaches zero at the Tsirelson endpoint.

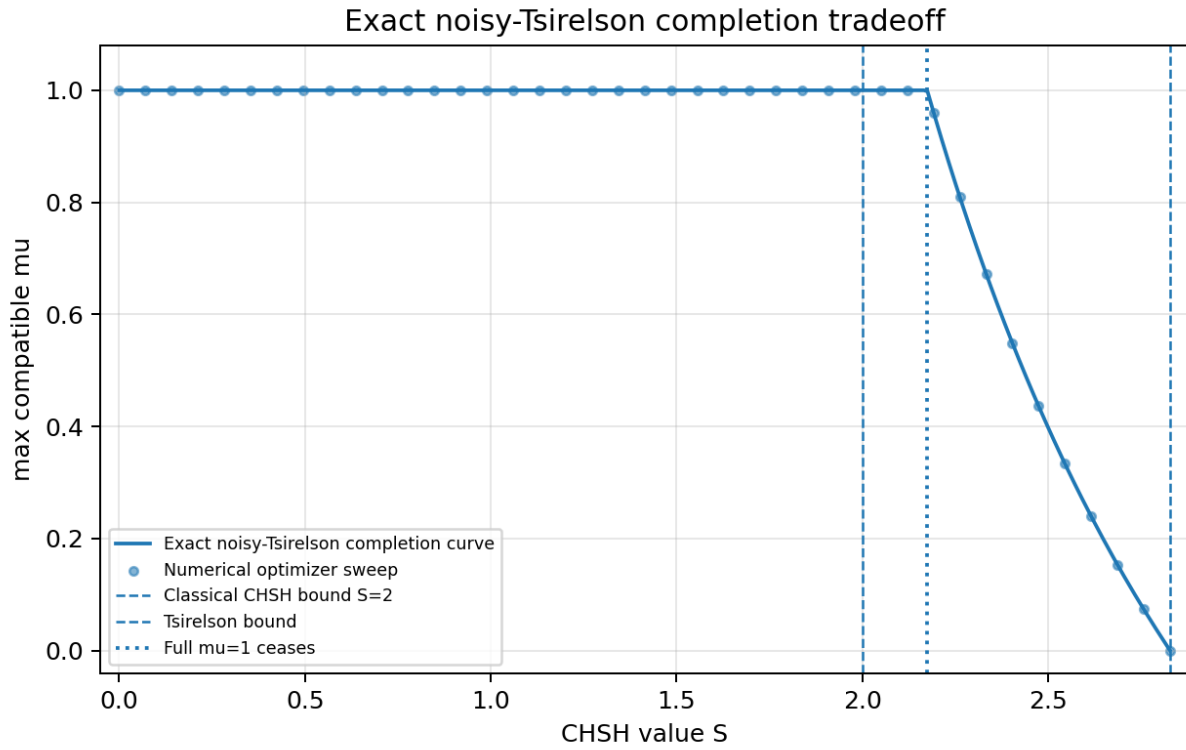


Figure 1. Exact noisy-Tsirelson completion tradeoff. The solid curve is the machine-checked formula for $\mu_{\max}$; points are the numerical optimizer sweep. Vertical lines mark the classical CHSH bound, the point where the full $\mu=1$ cap ceases, and the Tsirelson endpoint.

6.1 Representative values

visibility v	$x=v/\sqrt{2}$	CHSH $S=4x$	$t=\sqrt{1-x^2}-x$	$\mu_{\max}$
0.00	0.000000	0.000000	1.000000	1.000000
0.50	0.353553	1.414214	0.581861	1.000000
0.70	0.494975	1.979899	0.373933	1.000000
0.75	0.530330	2.121320	0.317461	1.000000
0.80	0.565685	2.262742	0.258936	0.809174
0.85	0.601041	2.404163	0.198178	0.548589
0.90	0.636396	2.545584	0.134966	0.333250
0.95	0.671751	2.687006	0.069025	0.152964
1.00	0.707107	2.828427	0.000000	0.000000

The exact drop from $\mu=1$ begins when $\sqrt{1-x^2}-x=x^2$, numerically $x \approx 0.543689$ and $S \approx 2.174756$. Thus full geometric consistency persists slightly beyond the classical CHSH bound $S=2$ along this particular path. This reinforces the central interpretation: Bell nonlocality and geometry are not simply mutually exclusive, but maximal nonlocality forces a collapse of uniform relational geometry.

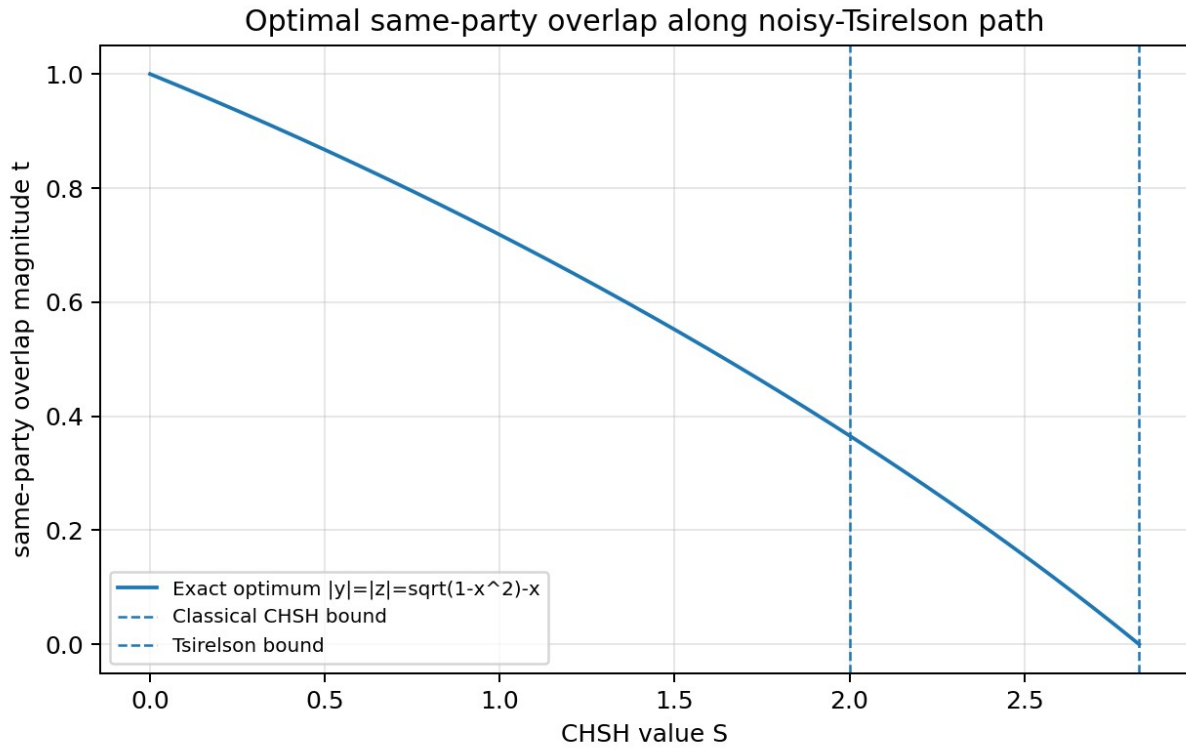


Figure 2. Optimal same-party overlap t along the noisy-Tsirelson path. As the CHSH value approaches $2\sqrt{2}$, the same-party overlaps required for a PSD completion go to zero, reproducing the orthogonality mechanism behind the general obstruction theorem.

7. Interpretation of the Exact Finding

The exact curve clarifies the physical interpretation. Geometry and Bell nonlocality are not enemies in general. Along the noisy-Tsirelson path, weak and even mildly Bell-violating correlations can still admit a fully geometrically consistent PSD completion. However, as the correlations approach the maximal quantum CHSH pattern, the best compatible value of μ is squeezed continuously to zero.

In plain language: ordinary geometry means relationships compose. If A is close to B and B is close to C, then A should retain some relation to C. The Tsirelson-optimal Bell pattern requires a special orthogonality structure in which certain same-party relations vanish while cross-party correlations remain large. That is precisely the pattern that breaks uniform geometric transitivity.

The theorem therefore supports a spectrum view. Local/geometric systems live near high μ . Strongly Bell-nonlocal systems require lower μ . The Tsirelson endpoint is the limiting case where the compatible uniform geometric consistency on the Bell sector disappears.

8. Exponential-Decay Bridge: When μ Arises Physically

The μ -submultiplicativity condition is not only an abstract algebraic constraint. It arises naturally when correlations are approximately exponential functions of metric distance. This is the standard qualitative form of correlations in many finite-correlation-length systems, diffusion kernels, transfer-matrix models, and gapped local systems.

Theorem 8.1 (Exponential decay implies μ -submultiplicativity). Let (X, d) be a metric space. Suppose that

for some $\xi > 0$ and $\epsilon \geq 0$, a kernel G satisfies

$$\exp(-\epsilon) \exp(-d(i,j)/\xi) \leq |G(i,j)| \leq \exp(\epsilon) \exp(-d(i,j)/\xi)$$

for all i, j . Then G satisfies μ -submultiplicativity with $\mu = \exp(-3\epsilon)$:

$$|G(i,k)| \geq \exp(-3\epsilon) |G(i,j)| |G(j,k)|.$$

8.1 Proof

By the lower approximation bound, $|G(i,k)| \geq \exp(-\epsilon) \exp(-d(i,k)/\xi)$. The triangle inequality gives $d(i,k) \leq d(i,j) + d(j,k)$, hence $\exp(-d(i,k)/\xi) \geq \exp(-(d(i,j) + d(j,k))/\xi)$. Therefore $|G(i,k)| \geq \exp(-\epsilon) \exp(-d(i,j)/\xi) \exp(-d(j,k)/\xi)$.

The upper approximation bounds imply $\exp(-d(i,j)/\xi) \geq \exp(-\epsilon) |G(i,j)|$ and $\exp(-d(j,k)/\xi) \geq \exp(-\epsilon) |G(j,k)|$. Substituting these two inequalities gives $|G(i,k)| \geq \exp(-3\epsilon) |G(i,j)| |G(j,k)|$, as claimed.

Corollary 8.2 (Exact exponential metric kernels). If $|G(i,j)| = \exp(-d(i,j)/\xi)$ for a genuine metric d , then G is 1-submultiplicative. Thus exact exponential metric decay gives exact relational geometry in the sense of this paper.

8.2 Physical interpretation

The bridge theorem identifies a broad physical regime where positive μ should be expected: whenever correlations decay approximately exponentially with metric distance. In that regime, $-\log|G|$ approximately recovers the underlying metric up to a controlled additive defect. The parameter ϵ controls the distortion from a perfect exponential metric kernel, and $\mu = \exp(-3\epsilon)$ quantifies how much relational transitivity survives.

This gives the Bell-sector obstruction a physical reading. Correlations that are strongly compatible with an exponential metric geometry cannot simultaneously realize the exact Tsirelson pattern on the same uniformly constrained Bell sector. Weak and moderate nonlocality may coexist with positive μ , but the Tsirelson endpoint forces the compatible geometric parameter to vanish.

9. Relation to the Full $C^*(\mu)$ Problem

Theorem 5.1 does not solve the full optimization $C^*(\mu)$, which asks for the largest possible CHSH value over all normalized PSD kernels satisfying μ -submultiplicativity. Instead, it solves the observable PSD-completion problem exactly on the canonical one-parameter noisy-Tsirelson path. This is still a substantial strengthening because it gives a closed-form tradeoff on the standard route to maximal quantum CHSH violation.

The remaining global problem is to characterize $C^*(\mu)$ over all feasible four-point Gram matrices, including asymmetric extremizers. The exact noisy-Tsirelson theorem provides a benchmark lower-dimensional solution and a test case for future SDP, KKT, or theorem-engine analyses.

10. Next Theorem and Empirical Targets

- Prove a robust inverse theorem: if $S \geq 2\sqrt{2} - \epsilon$, then $\mu_{\max}(E) \leq f(\epsilon)$ with $f(\epsilon) \rightarrow 0$.
- Derive certified upper and lower bounds for the full $C^*(\mu)$ curve over arbitrary Bell-sector PSD kernels.
- Apply the $\mu_{\max}(E)$ completion diagnostic to real Bell-test data, where only cross-correlators are observed.
- Compute $\mu(G)$ directly in simulated local spin-chain, diffusion, and tensor-network correlation kernels to test where geometric consistency naturally arises.

11. Conclusion

The revised framework now contains five layers: a general machine-checked obstruction theorem, an observable invariant $\mu_{\max}(E)$ for Bell data, an exact machine-checked completion curve for the noisy-Tsirelson family, an exponential-decay bridge theorem explaining when positive μ arises physically, and numerical visualizations that agree with the closed form. The main conceptual conclusion is sharpened: geometry and Bell nonlocality can coexist over a range of regimes, but maximal Bell nonlocality forces a break in uniform relational geometry. The full $C^*(\mu)$ curve remains open, but the canonical path to Tsirelson saturation is now solved exactly.

References

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Lean artifact: `RequestProject/NoisyTsirelsonCompletion.lean`, machine-checked exact noisy-Tsirelson completion theorem.

Bridge theorem added in this draft: approximate exponential metric decay implies μ -submultiplicativity with $\mu = \exp(-3 \epsilon)$.